DONISHA SMITH

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Research software engineer specializing in scientific computing, HPC pipeline automation, and open-source tool development

EDUCATION

Florida International University

PhD in Cognitive Neuroscience (M.Sc. en route Aug 2022) — Aug 2025

B.Sc. in Biological Sciences (Cum Laude; QBIC track), Honors College — Apr 2018

EXPERIENCE

Postdoctoral Researcher, Johns Hopkins University School of Medicine — Oct 2025–Present

Advisor: Dr. Kristin Bigos

- Developed a custom BIDS converter for multi-session pharmacological fMRI data lacking source DICOM/JSONs, reconstructing metadata from image headers and vendor documentation to enable integration with modern analysis pipelines (GitHub)
- Established the lab's first scalable neuroimaging infrastructure by migrating local GUI-based workflows to reproducible, containerized HPC pipelines using Apptainer with version-pinned neuroimaging tools (e.g., fMRIPrep, AFNI, FSL)
- Built an end-to-end fMRI analysis pipeline using SLURM job arrays and dependencies to parallelize and chain first-level GLM/gPPI modeling, parametric and nonparametric group inference, MNI region labeling, and subject-level cluster beta extraction (GitHub)

Graduate Researcher, Florida International University — Aug 2019–Aug 2025

Advisor: Dr. Angela Laird

- Developed and published a k-means clustering pipeline to characterize dynamic brain state transitions in longitudinal fMRI data, identifying distinct transition patterns across timepoints
- Applied mixed effects, structural equation, and latent profile models to multi-session fMRI and behavioral data to identify dissociable brain-behavioral profiles, contributing to two peer-reviewed publications
- Analyzed staff demographic data for the ABCD Study consortium, implementing hypothesis testing and data visualization to inform decisions for a multi-site research task force (GitHub)

Medical Laboratory Scientist I, Florida Department of Health — Jan 2019–Jul 2019

- Conducted confirmatory diagnostic testing for HIV and syphilis for public health surveillance programs

OPEN-SOURCE PROJECTS

BIDSAid | GitHub | Docs — Oct 2025–Present

- Developed a Python package for post-hoc BIDS conversion of NIfTI datasets lacking source DICOMs, reconstructing JSON sidecars with acquisition metadata derived from image headers
- Built configurable parsers for Presentation and E-Prime behavioral logs, automating timing and accuracy extraction for block and event-related fMRI designs into BIDS-compliant events files
- Implemented cross-platform testing across Python 3.10–3.14 on Linux, macOS, and Windows, maintaining >90% code coverage

NeuroCAPs | GitHub | Docs | Google Colab | JOSS Paper | JHU OSPO Catalog — Jan 2024–Present

- Built a Python package for brain state classification using k-means, with multiprocessing for multi-subject timeseries extraction
- Incorporated clustering validation (e.g., Davies-Bouldin, Silhouette), KD-tree interpolation for volumetric-to-surface mapping, temporal dynamics metrics (e.g., transition probabilities), and integrated visualizations (e.g., surface maps, network alignment)
- Deployed via Docker with headless VTK rendering and CLI/Jupyter interfaces; maintained >90% test coverage across several OS platforms using pytest and GitHub Actions

vswift | GitHub — Apr 2023–Present

- Created an R package for ML model evaluation, implementing custom stratified sampling, cross-validation, and nested cross-validation for hyperparameter tuning, with parallel processing support at the fold-level
- Designed a unified interface supporting classification algorithms (e.g., Regularized Logistic Regression, SVM, XGBoost, Neural Networks) while preserving algorithm-specific parameter tuning
- Incorporated automated missing data imputation and multi-metric performance assessment (e.g., precision, recall, F1) with CLI output and visualizations, including ROC and Precision-Recall curves with AUC scores

PUBLICATIONS

[1] **Smith, D. D.**, Bartley, J. E., Peraza, J. A., Bottenhorn, K. L., Nomi, J. S., Uddin, L. Q., Riedel, M. C., Salo, T., Laird, R. W., Pruden, S. M., Sutherland, M. T., Brewe, E., & Laird, A. R. (2025). Dynamic reconfiguration of brain coactivation states associated with active and lecture-based learning of university physics. Npj Science of Learning, 10(1), 55. https://doi.org/10.1038/s41539-025-00348-9

[2] **Smith, D**. (2025). NeuroCAPs: A Python Package for Performing Co-Activation Patterns Analyses on Resting-State and Task-Based fMRI Data. Journal of Open Source Software, 10(112), 8196. https://doi.org/10.21105/joss.08196

[3] Pintos Lobo, R., Peraza, J. A., Salo, T., Meca, A., **Smith, D. D.**, Feeney, K. E., Schmarder, K. M., Sutherland, M. T., Gonzalez, R., Musser, E. D., & Laird, A. R. (2025). Social profiles among youth with attention-deficit/hyperactivity disorder (ADHD): Evidence from the ABCD study. Developmental Cognitive Neuroscience, 75, 101591. https://doi.org/10.1016/j.dcn.2025.101591

[4] **Smith D. D.**, Meca A, Bartley JE, Riedel MC, Salo T, Peraza JA, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird AR (2023). Task-based attentional and default mode connectivity associated with science and math anxiety profiles among university physics students. Trends in Neuroscience and Education. https://doi.org/10.1016/j.tine.2023.100204

[5] Lobo, R. P., Bottenhorn, K. L., Riedel, M. C., Toma, A. I., Hare, M. M., **Smith, D. D.**, Moor, A. C., Cowan, I. K., Valdes, J. A., Bartley, J. E., Salo, T., Boeving, E. R., Pankey, B., Sutherland, M. T., Musser, E. D., & Laird, A. R. (2022). Neural systems underlying RDoC social constructs: An activation likelihood estimation meta-analysis. Neuroscience & Biobehavioral Reviews. https://doi.org/10.1016/j.neubiorev.2022.104971

PRESENTATIONS

[1] **Smith D. D.**, Bartley JE, Peraza JA, Riedel MC, Salo T, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird A.R. Longitudinal Changes in Dynamic Functional Connectivity Associated with Physics Learning. Presented on May 3, 2024, at the Florida Consortium on the Neurobiology of Cognition; Miami, Florida.

[2] **Smith D. D.**, Bartley JE, Peraza JA, Riedel MC, Salo T, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird A.R. Longitudinal Changes in Dynamic Functional Connectivity Associated with Physics Learning. Accepted at the 2024 Graduate Student Appreciation Week at Florida International University; Miami, Florida.

[3] **Smith D. D.**, Meca A, Bartley JE, Riedel MC, Salo T, Peraza JA, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird A.R. Task-based attention & default mode connectivity linked to STEM anxiety in university students. Presented on July 23, 2023, at the 29th annual meeting of the Organization for Human Brain Mapping; Montréal, Canada.

TEACHING & WORKSHOPS

Introduction to R Workshop | GitHub — Feb 2023

- Designed and delivered an introductory workshop on R programming and Tidyverse for the Diversity, Equity, and Inclusion program, instructing 10+ graduate students on data analysis, manipulation, and visualization

PEER REVIEW

Journal of Open Source Software — Aug 2025–Present

npj Science of Learning — Jul 2025–Present

FELLOWSHIPS, HONORS & SCHOLARSHIPS

Dissertation Year Fellowship ($20,000 across two semesters) — Dec 2024–Jun 2025

DEI Doctoral Fellowship ($3,000 over three semesters) — Aug 2022–Aug 2023

Recipient of Undergraduate NIGMS RISE Fellowship — Jan 2017

FIU Ambassador Scholarship (full-tuition merit scholarship, 2014 cohort) — Aug 2014–May 2018

QBIC Scholarship ($1,000 per semester) — Aug 2014–May 2018

Florida Academic Scholars Award (highest state merit scholarship tier) — Aug 2014–May 2018

TECHNICAL SKILLS

Programming Languages: Python, R, Shell Scripting (Bash), SQL

Computing & Development: High-Performance Computing, CI/CD (Git, GitHub Actions, Docker, Singularity/Apptainer)

Neuroimaging Tools: Nilearn, fMRIPrep, AFNI, FSL, E-PRIME, Presentation, PsychoPy